

WORKING WITH DENTAL ACRYLICS

Best Practices

Background

Dental acrylic materials are widely used during rodent implantation procedures in animal research. The dental acrylic is used to secure the implantation device on the rodent's skull.

A few milligrams (2-3 mg) of powdered dental acrylic are mixed with a few milliliters (0.5 - 10 mL) of the liquid acrylics to get the powders wet. The mixture is typically applied around the implantation device right after mixing and cures within 2-3 minutes. The dental acrylic mixture helps create a strong head cap fixed to the rodent's skull.

The use of dental acrylic materials may pose hazards, so it is essential that safety control measures are implemented.



Dental Acrylic Materials

Dental Acrylic: Hazards

- The dental acrylic powders that may cause allergic skin reactions and eye irritation including: zirconium oxide, 2-propenoic acid, 2-methyl-methyl ester, homopolymer (80-90 %w/w), diethyl phthalate (10-20% w/w), benzoyl peroxide (1-5% w/w), and titanium dioxide (0-1% w/w).
- The dental acrylic solution often contains methyl methacrylate. Methyl methacrylate can cause skin irritation, eye irritation, allergic skin reaction, and respiratory irritation. In addition, methyl methacrylate is a highly flammable liquid and vapor.
- Catalyst solution containing 2-hydroxyethyl methacrylate and can also be used for the dental acrylic mixture. The catalyst solution can cause skin irritation, allergic skin reaction, and eye irritation.

Safe, Best Practices

- Use an active waste gas scavenging system (e.g., chemical fume hood, snorkel trunk, EVAC-4, VetEquip cube, RWD) when conducting a rodent implantation procedure to minimize the inhalable risk of dental acrylics. Since isoflurane is used to induce and maintain anesthesia during rodent implantation procedures, an active waste gas scavenging system is required to ensure safe work with isoflurane. Further information on safe work with isoflurane can be found [here](#). The active waste gas scavenging system which is ducted to building exhaust system or connected to an activated charcoal canister reduces personnel exposure to both isoflurane and dental acrylics materials.
- If there is a chemical fume hood nearby, mix the dental acrylic materials in a chemical fume hood, if feasible to further minimize the risk of dental acrylics exposure. Given small quantities of dental acrylics

used and short duration of use, the risk of dental acrylics inhalation exposure is very low if the mixing is done on a surgical station.

- Keep the flammable dental acrylic liquid solution away from heat sources.
- Keep all chemical containers tightly closed after each use.
- Remove gloves carefully and wash hands thoroughly after glove removal to avoid skin exposure.
- Wear a lab coat, chemical-resistant gloves, and safety glasses when handling the dental acrylic materials.

References

[C&B Metabond Powder Safety Data Sheet](#)

[C&B Metabond Quick Base Solution Safety Data Sheet](#)

[Amalgambond Catalyst Safety Data Sheet](#)

[Lang Ortho-Jet 1306 Safety Data Sheet](#)

[Lang Ortho-Jet 1330 Safety Data Sheet](#)

Compton et al. (2020). Masks for the Reduction of Methyl Methacrylate Vapor Inhalation. The Iowa Orthopaedic Journal 40(1): 191–193. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7368533/#>